APPLE GOOGLE

Twitter Sentiment Analysis

Apple & Google Products

Phase 4 Group 4 Project | NLP Sentiment Classification

9,093	3	6+
Tweets Analyzed	Sentiment Classes	Models Compared

Presentation Overview

A plain-language guide to our findings

01 **Business Understanding**
What problem are we solving and who benefits?

02 **Data Understanding**
The 9,093 tweets we analyzed

03 **Modeling**
How our Model learned to detect sentiment

04 **Evaluation**
How well our model performs

05 **Recommendations & Next Steps**
What this means for your business

Business Understanding

The Problem We're Solving

Every hour, Apple and Google receive thousands of social media mentions. Manually reading and categorizing all of them is impossible — yet knowing whether public opinion is positive, negative, or neutral is critical for protecting brand reputation, guiding product development, and informing marketing strategy.

Who Benefits?

Marketing Teams

Monitor product launch sentiment in real-time and amplify positive buzz automatically

Product Development

Identify recurring complaints to prioritize engineering fixes on the product roadmap

Customer Support

Alert teams when negative sentiment spikes — proactively handle issues before they escalate

Executive / PR

Track brand health over time with data-driven sentiment dashboards for leadership

Data Understanding

What data did we use and what does it tell us?

About the Dataset

- 9,093 real tweets from Twitter
- Collected at SXSW Tech Conference 2011
- Source: CrowdFlower via data.world
- Human experts labeled each tweet
- Covers Apple & Google products

Sentiment Breakdown

Challenge: Only 6% of tweets are negative — the hardest class to detect!

Data Limitations (Known Upfront)

2011 Only

Products & language have changed since then

SXSW Bias

Tech-conference crowd, not everyday users

Ambiguity

Some tweets are hard even for human raters

How the model learned to detect sentiments

1

Clean the Text

Remove website links, @usernames & hashtag symbols. Keep only the words that carry meaning.

2

Convert Words to Numbers

Every word gets a score. Rare but important words (like 'crashed') score higher than common words.

3

Train the Model

We fed thousands of labeled tweets to 6+ algorithms. They learn which words indicate each sentiment.

4

Classify New Tweets

Given any new tweet, the model instantly labels it Positive, Negative, or Neutral.

Think of it like teaching someone by example: show them 9,000 labeled tweets, and they learn the patterns. Our best model does this with ~85% accuracy.

Models We Tested

Algorithm	Plain-Language Explanation	Performance
Logistic Regression	Weighs each word's evidence for each sentiment — clear, transparent, and fast	**★ WINNER — Best overall**
Linear SVC	Draws the sharpest possible boundary between sentiment groups in the data	Very close 2nd place
Naive Bayes	Counts word frequencies — the classic approach used in email spam filters	Decent on common words
Random Forest	Builds hundreds of decision trees and takes a majority vote — like a committee	Good, not best for text
XGBoost	Learns by correcting past mistakes, step by step — a powerful boosting technique	Strong non-linear model
spaCy Embeddings	Understands word meaning: 'great' and 'excellent' are treated as similar	TF-IDF won for this task

How Well Does It Work?

Evaluation — Understanding Model Performance Without Jargon

Performance at a Glance

~85%

Overall Score

Across all 3 sentiment types (Weighted F1)

~89%

Positive Tweets

Correctly identified as positive

~87%

Neutral Tweets

Correctly identified as neutral

~62%

Negative Tweets

Hardest — only 6% of training data!

Why not just use % accuracy?

A model that always guesses 'Neutral' scores 60% accuracy — but is completely useless! Our F1-Score metric accounts for this, ensuring all 3 classes matter equally.

Where Our Model Struggles:

Neutral ↔ Positive

Mild positive language gets misread

Negative Detection

Too few negative examples to learn from

Sarcasm & Irony

"Great, it crashed again" fools the Model

Business Recommendations

What this model means for your teams — in plain terms

Marketing Teams

Use the model to flag positive tweets in real-time during product launches. Amplify them for reach. Set automated alerts when positive volume drops unexpectedly.

→ **Faster, data-driven campaign response**

Product Development

Mine the top negative-sentiment keywords — 'crash', 'battery', 'slow' — to guide your engineering roadmap. These words are direct customer feedback at scale.

→ **Product fixes grounded in real user voice**

Customer Support

Deploy negative-sentiment spike alerts so support teams can respond before issues go viral. Route high-negative-volume topics to senior agents immediately.

→ **Proactive CX — prevent PR crises**

Executive / PR

Build a live brand health dashboard using this model. Track sentiment scores week-over-week and compare Apple vs. Google perception over time.

→ **Executive-grade brand intelligence**

Next Steps

How we make this even more powerful

01

Upgrade to Transformer Based Models

Use next-gen models that understand full sentence context — detecting sarcasm like a human would.

02

Retrain on Current Tweets

Our data is from 2011. New training data will ensure the model understands modern words and current products.

03

Build a Live Dashboard

Deploy as a real-time API connected to the Twitter/X stream, feeding results into a Tableau dashboard.

04

Aspect-Level Analysis

Instead of one label per tweet, pinpoint which features are praised or criticized (camera, battery, software).

05

Expand to More Brands

Extend to competitor brands for a full side-by-side sentiment comparison across the smartphone market.

06

A/B Test Marketing Copy

Use positive-sentiment word patterns to craft marketing language that actually resonates with customers.

Thank You!

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- Logistic Regression best captures tweet sentiment
- Positive & Neutral: ~87–89% accuracy achieved
- Negative detection (62%) improves with more data
- Every prediction is fully explainable & transparent

Project Contributors

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